

Network Layer & Addressing – Question List by Marks

5 Marks Questions

Network Layer

1. Define the Network Layer.
2. List the functions of the Network Layer.
3. What is routing?
4. What is packet forwarding?
5. What is a router?
6. What is fragmentation?
7. What is reassembly?
8. What is congestion?
9. Differentiate between routing and forwarding.
10. What is TTL (Time To Live)?
11. What is a routing table?

Addressing

12. What is an IP address?
13. What is logical addressing?
14. What is IPv4?
15. What is IPv6?
16. What is a subnet mask?
17. What is a network address?
18. What is a broadcast address?
19. What is a private IP address?
20. What is a public IP address?
21. What is NAT?
22. What is DHCP?
23. What is the default gateway?
24. What is ARP?

25. What is RARP?
 26. What is ICMP?
 27. What is Ping?
 28. What is Traceroute?
 29. What is Unicast communication?
 30. What is Broadcast communication?
 31. What is Multicast communication?
 32. What is Anycast communication?
 33. Write the features of IPv6.
 34. What is CIDR?
 35. What is Supernetting?
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10 Marks Questions

Network Layer

1. Explain the responsibilities of the Network Layer.
2. Explain routing in detail.
3. Explain packet forwarding with an example.
4. Describe the working of a router.
5. Explain fragmentation and reassembly.
6. Explain congestion and congestion control.
7. Explain the IPv4 packet format.
8. Differentiate between static routing and dynamic routing.
9. Explain routing tables and their importance.
10. Explain path determination in the Network Layer.

Addressing

11. Explain IPv4 addressing.
12. Explain IPv6 addressing.
13. Compare IPv4 and IPv6.

14. Explain the classes of IPv4 addresses.
 15. Explain subnet masks with examples.
 16. Explain subnetting with examples.
 17. Explain CIDR addressing.
 18. Explain Supernetting.
 19. Explain NAT and its working.
 20. Explain DHCP and its operation.
 21. Explain ARP and RARP protocols.
 22. Explain ICMP and its applications.
 23. Explain public and private IP addressing.
 24. Explain Unicast, Broadcast and Multicast communication.
 25. Explain Anycast communication with examples.
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15 Marks Questions

1. Explain the Network Layer in detail with its functions and diagram.
2. Describe the IPv4 packet format with a neat diagram.
3. Explain IPv6 addressing in detail and compare it with IPv4.
4. Explain IP addressing and the classification of IPv4 addresses.
5. Explain subnetting with suitable numerical examples.
6. Explain CIDR and Supernetting in detail.
7. Explain NAT with diagram, working, advantages and disadvantages.
8. Explain routing and routing algorithms in detail.
9. Explain static routing and dynamic routing with comparison.
10. Explain ARP, RARP and ICMP protocols in detail.
11. Explain DHCP and its message exchange process.
12. Explain logical addressing and its importance in networking.
13. Explain fragmentation and reassembly with suitable examples.
14. Explain congestion control techniques in the Network Layer.

15. Explain packet forwarding and routing table lookup process.
16. Compare IPv4 and IPv6 in detail.
17. Explain different types of IP communication: Unicast, Broadcast, Multicast and Anycast.
18. Explain the working of routers in a network.
19. Discuss public IP addressing and private IP addressing in detail.
20. Explain the complete process of packet delivery from source to destination through the Network Layer.

Most Important Exam Questions (Frequently Asked)

15 Marks

1. Network Layer and its functions.
2. IPv4 Packet Format.
3. IPv6 and IPv4 comparison.
4. Subnetting.
5. NAT.
6. Routing and Routing Algorithms.
7. ARP, RARP and ICMP.
8. CIDR and Supernetting.

10 Marks

1. IPv4 Addressing.
2. Classes of IP Addresses.
3. DHCP.
4. Congestion Control.
5. Router and Packet Forwarding.

5 Marks

1. NAT.
2. ARP.
3. ICMP.
4. TTL.

5. DHCP.
6. Gateway.
7. Private and Public IP Address.
8. Broadcast Address.
9. Routing Table.
10. IPv6 Features.

Additional 5-Mark Questions

1. What is IPv4 address exhaustion?
2. What are the security limitations of IPv4?
3. What is routing inefficiency in IPv4?
4. What is a Dual Stack network?
5. What is IPv6 Tunneling?
6. What is IPv6 Translation?
7. What is the simplified IPv6 header?
8. Why was IPv6 introduced?

Additional 10-Mark Questions

1. Explain IPv4 address exhaustion and how IPv6 solves it.
2. Explain the limitations of IPv4.
3. Explain the simplified IPv6 header format.
4. Explain the Dual Stack transition mechanism.
5. Explain IPv6 Tunneling with examples.
6. Explain IPv6 Translation techniques.
7. Explain the need for IPv6 in modern networks.
8. Explain the security improvements in IPv6 over IPv4.
9. Explain routing efficiency improvements in IPv6.

Additional 15-Mark Questions

1. Discuss the limitations of IPv4 and the need for IPv6.
2. Explain IPv6 features, address format, and advantages over IPv4.
3. Describe the IPv6 header format and explain how it is simpler than the IPv4 header.
4. Explain IPv4-to-IPv6 transition mechanisms:
 - Dual Stack
 - Tunneling
 - Translation
5. Explain IPv6 addressing architecture and its benefits.
6. Compare IPv4 and IPv6 with respect to:
 - Address space
 - Security
 - Routing efficiency
 - Header structure
 - Configuration
7. Explain the challenges of IPv4 address exhaustion and the solutions provided by IPv6.